Intelligent Systems

| *ReviewPulse*

A multi-domain Sentiment Classifier

INTELLIGENT SYSTEMS (ISY503)
ASSESSMENT 1

PROF DR. NANDINI SIDNAL

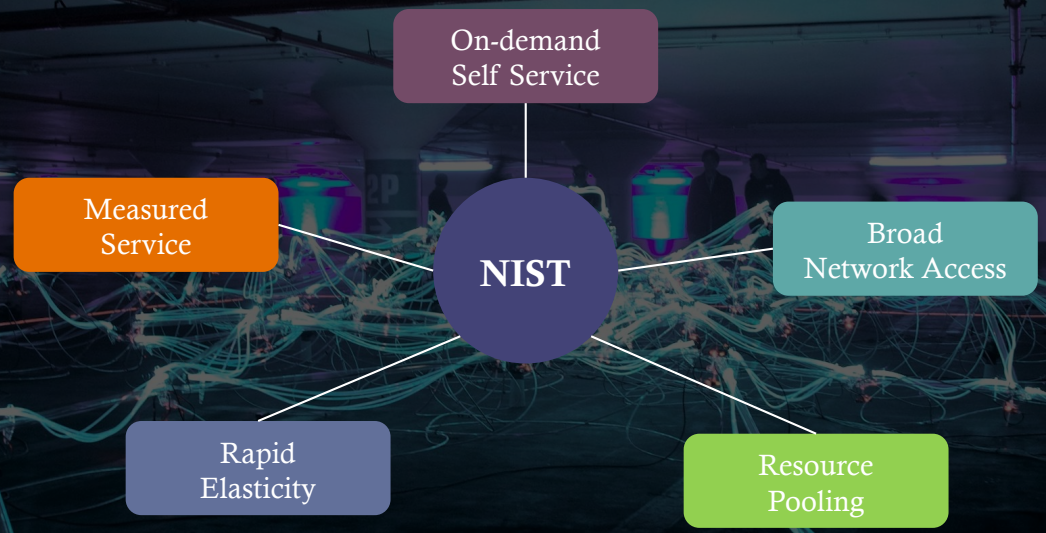
LUIS G. B. A. FARIA – ID A00187785
SAMIRAN SHRESTHA – ID A00106473
VICTOR J. D. MENESES – ID A00179705

May, 2026

TORRENS UNIVERSITY AUSTRALIA

PROBLEM STATEMENT

On-demand delivery of compute, storage and services over the internet (Mell & Grance, 2011).



Key shift: from **owning capacity → renting capability**

TORRENS UNIVERSITY AUSTRALIA

DATASET

B1 – Cost Efficiency

Traditional IT (CapEx)	Cloud – ABC (OpEx)
Large upfront hardware purchase	**No upfront servers**
Idle capacity during low demand	**Pay per compute hour + GB-month**
Full ops team required	**Provider manages infrastructure**
Fixed costs regardless of usage	**~80% reduction in start-up IT costs**

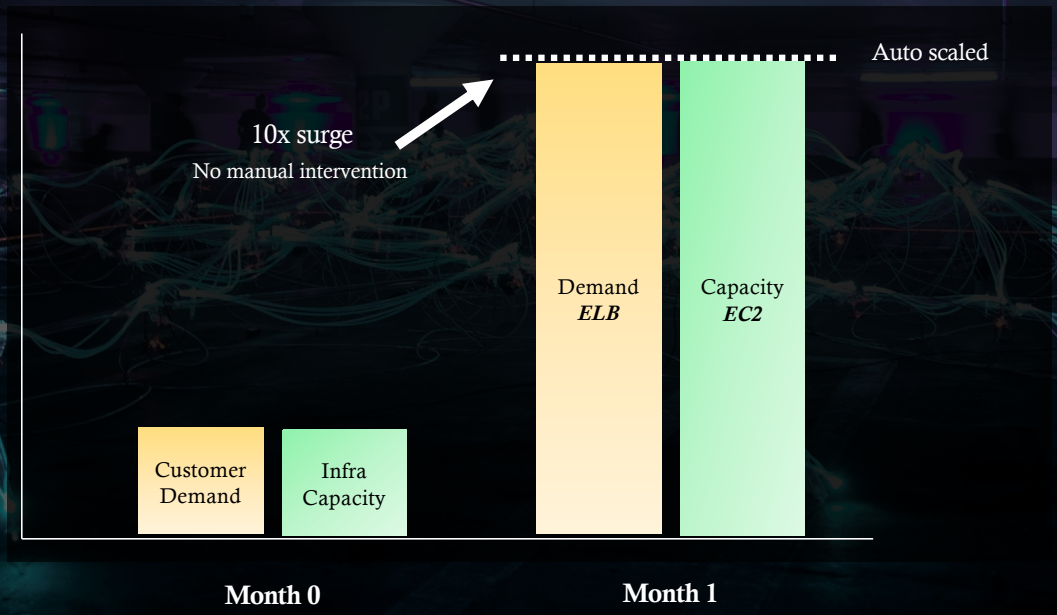
COST SHIFT

TORRENS UNIVERSITY AUSTRALIA

PREPROCESSING PIPELINE

B2 & B3 – Scale without Limit & Less Overhead

Rapid Elasticity: ABC's 10x Growth absorbed by Auto Scaling



TORRENS UNIVERSITY AUSTRALIA

ARCHITECTURE OVERVIEW

Challenge	*Example*	*Mitigation*
Security & Privacy	PII + payment data	IAM least-privilege, MFA, encryption, shared responsibility model
Cost Volatility	Pay-as-you-go can spiral without guardrails	FinOps: budget alerts, resource tagging, rightsizing, reserved pricing
Vendor Lock-in + Skills Gap	Migration cost + mindset shift	Portability-first (containers, standard DBs), targeted cloud upskilling

TRAINING

Cloud Service Models

IaaS
Compute + Config
Flexibility

PaaS
Managed DB, load balancing,
serverless

SaaS
Limited customization

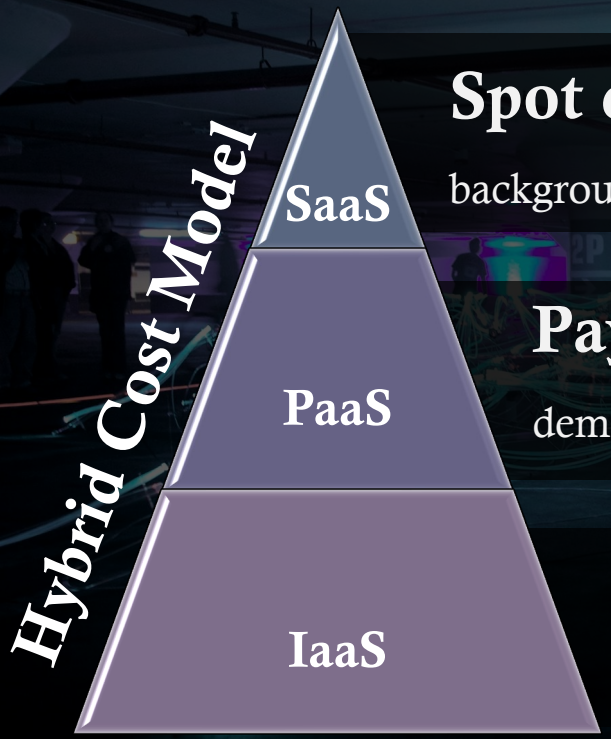
Cloud Deployment Models

Private cloud
High cost, limited elasticity

Hybrid cloud
Premature complexity

Public cloud
Scalable, automated,
cost-effective

RESULTS



Spot capacity

background jobs + analytics pipelines

Pay-as-you-go

demand spikes via Auto Scaling

Reserved

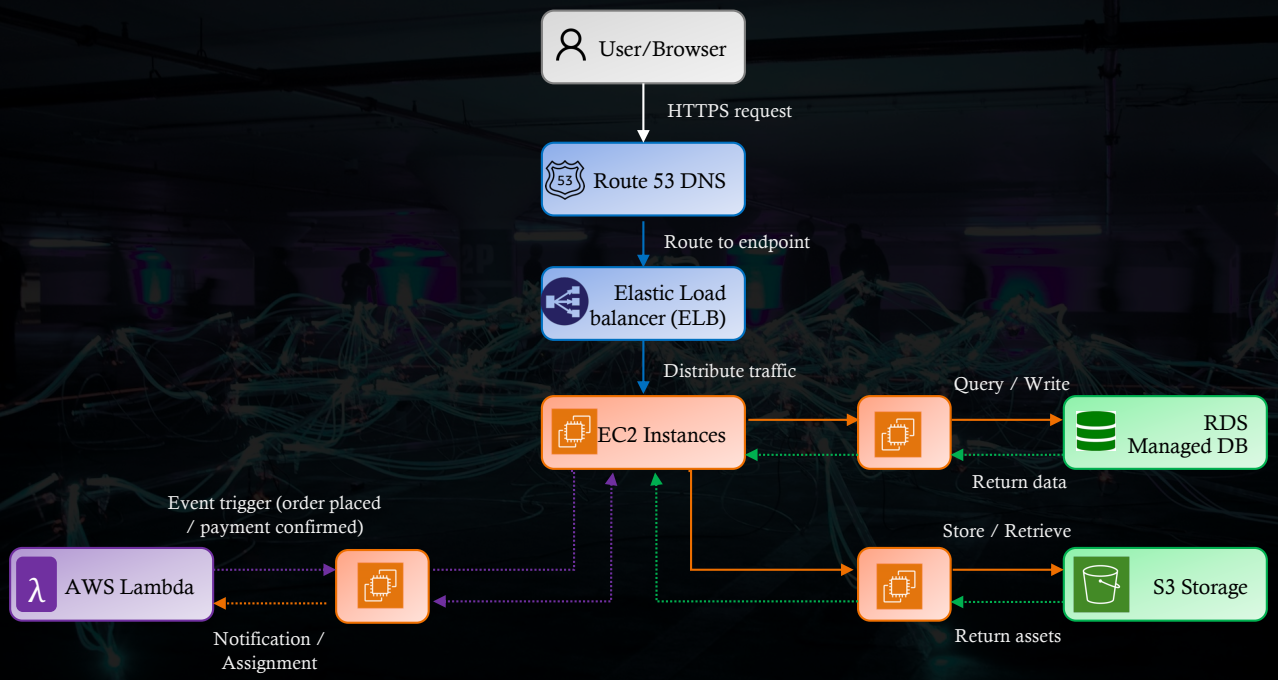
stable tiers (web/app servers, databases)

TORRENS UNIVERSITY AUSTRALIA

Error Analysis

Provider	Ecosystem Fit	Load Balancing	Serverless	ABC Alignment
aws	Broadest managed-service catalogue	ELB – native Route53 integration	Lambda – event-driven, zero idle cost	Best fit – Route 53 already in stack
Azure	Microsoft / enterprise-aligned	Application gateway – extra config needed	Azure Functions – strong but separate ecosystem	No existing Azure signals in ABC
Google Cloud	Analytics and ML-first	Cloud load balancing – GKE-oriented	Cloud Run / Functions – container first	No analytics-heavy workloads in ABC now

Live Demo



TORRENS UNIVERSITY AUSTRALIA

ETHICS & LIMITATIONS

Public cloud | IaaS+PaaS | Hybrid cost model | AWS

Key Outcomes Unlocked:

- 80% cost reduction (measured service)
- 10x surge absorbed (rapid elasticity)
- Zero-headcount scaling (resource pooling + Auto Scaling)
- Workflow automation (Lambda event triggers)

FUTURE WORK

Public cloud | IaaS+PaaS | Hybrid cost model | AWS

Key Outcomes Unlocked:

- 80% cost reduction (measured service)
- 10x surge absorbed (rapid elasticity)
- Zero-headcount scaling (resource pooling + Auto Scaling)
- Workflow automation (Lambda event triggers)

SUMMARY / QUESTIONS

Public cloud | IaaS+PaaS | Hybrid cost model | AWS

Key Outcomes Unlocked:

- 80% cost reduction (measured service)
- 10x surge absorbed (rapid elasticity)
- Zero-headcount scaling (resource pooling + Auto Scaling)
- Workflow automation (Lambda event triggers)

References

Amazon Web Services. (n.d.-a). *AWS Well-Architected Framework.* https://aws.amazon.com/architecture/well-architected/

Amazon Web Services. (n.d.-b). *AWS Pricing.* https://aws.amazon.com/pricing/

Bittok, T. (2022, January 22). *Cloud total cost of ownership.* LinkedIn Pulse. https://www.linkedin.com/pulse/cloud-total-cost-ownership-theophilus-bittok-/

Carpenter, T. (2020, May 27). *AWS Certified Solutions Architect – Associate (SAA-C02) Cert Prep 1: Cloud services overview*. LinkedIn Learning. https://www.linkedin.com/learning/aws-certified-solutions-architect-associate-saa-c02-cert-prep-1-cloud-services-overview/benefits-of-cloud-computing

Eliaçık, E. (2022). *Pros and cons of cloud computing.* Dataconomy. https://dataconomy.com/2022/05/pros-and-cons-of-cloud-computing-2022/

IBM. (n.d.-a). *SaaS, PaaS, IaaS explained.* IBM Cloud. https://www.ibm.com/think/topics/iaas-paas-saas

IBM. (n.d.-b). *What is a public cloud?* IBM. https://www.ibm.com/think/topics/public-cloud

IBM. (n.d.-c). *What is private cloud?* IBM. https://www.ibm.com/think/topics/private-cloud

Linthicum, D. (2021, May 25). *Learning cloud computing: Core concepts.* LinkedIn Learning. https://www.linkedin.com/learning/learning-cloud-computing-core-concepts-13710481/

Manvi, S., & Shyam, G. K. (2021). *Cloud computing: Concepts and technologies* (Chapter 4). CRC Press. https://learning-oreilly-com.torrens.idm.oclc.org/library/view/cloud-computing/9781000338058/

McHaney, R. (2021). *Cloud technologies: An overview of cloud computing technologies for managers.* Wiley. https://ieeexplore-ieee-org.torrens.idm.oclc.org/servlet/opac?bknumber=9820907

Mell, P., & Grance, T. (2011). *The NIST definition of cloud computing* (Special Publication 800-145). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-145

Naved, M., Sanchez, D. T., Dela Cruz, A. P., Jr., Peconcillo, L. B., Jr., Peteros, E. D., & Tenerife, J. J. L. (2022). Identifying the role of cloud computing technology in management of educational institutions. *Materials Today: Proceedings, 51*(8), 2309–2312. https://doi.org/10.1016/j.matpr.2021.11.414

Nishimura, H. (2022, August 30). *Introduction to AWS for non-engineers: 1 cloud concepts*. LinkedIn Learning. https://www.linkedin.com/learning/introduction-to-aws-for-non-engineers-1-cloud-concepts-2/

Shore, J. (2020). *Cybersecurity with cloud computing: Service models*. LinkedIn Learning. https://www.linkedin.com/learning/cybersecurity-with-cloud-computing-2/

Thank you!

INTELLIGENT SYSTEMS (ISY503)
ASSESSMENT 1

PROF DR. NANDINI SIDNAL

LUIS G. B. A. FARIA – ID A00187785
SAMIRAN SHRESTHA – ID A00106473
VICTOR J. D. MENESES – ID A00179705

May, 2026

TORRENS UNIVERSITY AUSTRALIA